

XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 3: Kinematics - Motion in 2D and 3D

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In this lecture, you'll learn…

- how to use vectors to represent the position and velocity of a particle in two or three dimensions.
- how to find the vector acceleration of a particle, and how to interpret the components of acceleration parallel to and perpendicular to a particle's path.
- how to solve problems that involve the curved path followed by a projectile.
- how to analyze motion in a circular path, with either constant speed or varying speed.
- how to relate the velocities of a moving object as seen from two different frames of reference.

v. 解释. 口译

adj. 垂直的, 成直角的

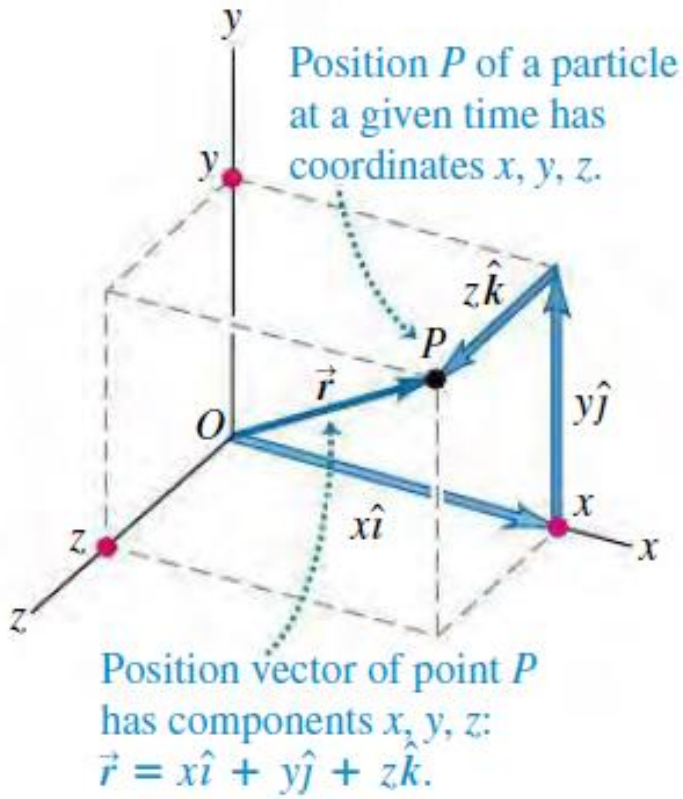
Physical Quantities in 2D & 3D Motion

- Physical quantities studied in 2D & 3D Motion 物理量
  - ➢ Displacement/Position Vector: ***r*** or ***r***(t)
  - ➢ Velocity: ***v***(***r***, t) or ***v***(***r***), or ***v***(t)
  - ➢ Acceleration: ***a***(***r***, t) or ***a***(***r***), or ***a***(t)
- The position vector from the origin to point *P* has components *x*, *y*, and *z*.

Position vector of a particle at a given instant $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ Coordinates of particle's position

Unit vectors in *x*-, *y*-, and *z*-directions

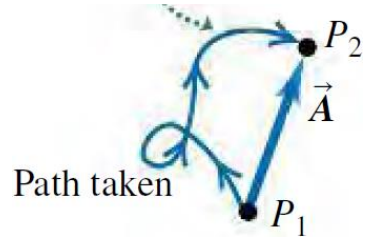
(3.1)



Displacement and Position Vector

- Displacement, as defined in Chapter 1, is a vector.
- The starting position at P_1 (x_1,y_1,z_1), and ending position at P_2 (x_2,y_2,z_2), so

$$\boldsymbol{r}_1 = x_1\boldsymbol{i} + y_1\boldsymbol{j} + z_1\boldsymbol{k}$$
$$\boldsymbol{r}_2 = x_2\boldsymbol{i} + y_2\boldsymbol{j} + z_2\boldsymbol{k}$$



and change of position, i.e. the displacement, can be written as:

$$\Delta\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1 = (x_2 - x_1)\,\boldsymbol{i} + (y_2 - y_1)\,\boldsymbol{j} + (z_2 - z_1)\,\boldsymbol{k}$$

- If the starting position at P_1 is (0, 0, 0) and ending position at P_2 is (x, y, z), the change in position or displacement can be written as:

$$\Delta\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$$

Velocity

- Average Velocity **vector**
  - ➢ We define the average velocity as the displacement ***Δr*** divided by the time interval *Δt* the particle moves between two points from P_1 to P_2:

Change in the particle's position vector

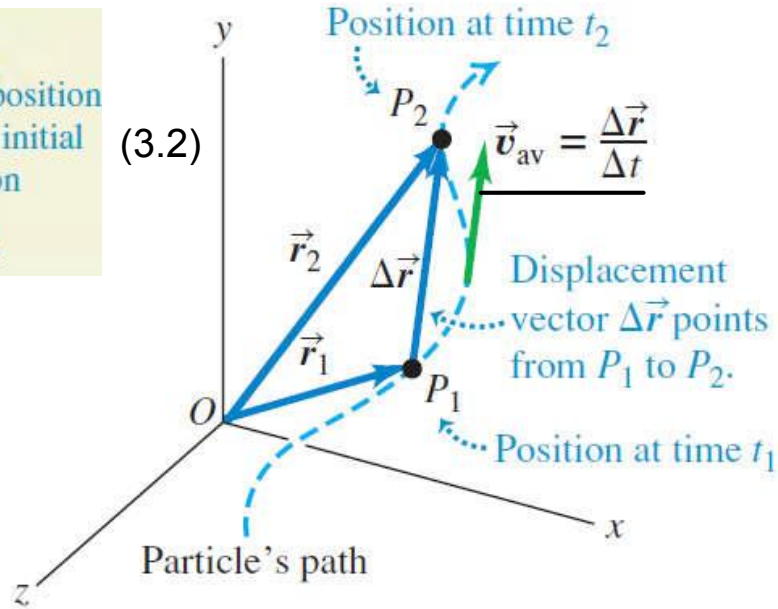
Average velocity **vector** of a particle during time interval from t_1 to t_2

$$\vec{v}_{av} = \frac{\Delta\vec{r}}{\Delta t} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} \quad (3.2)$$

Final position minus initial position

Time interval

Final time minus initial time



  - ➢ It has the same direction as the displacement.

平均速度向量

Velocity

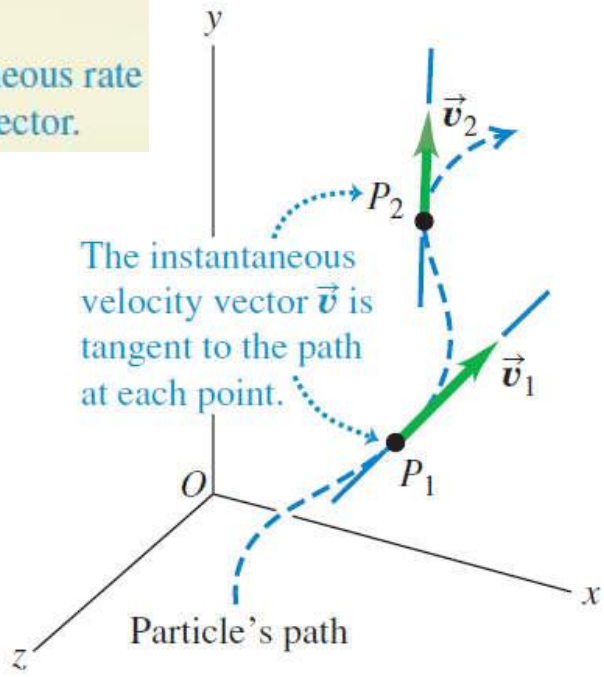
- Instantaneous Velocity can be expressed as

The **instantaneous velocity vector** of a particle ... $\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$ (3.3)

... equals the limit of its average velocity vector as the time interval approaches zero ...

... and equals the instantaneous rate of change of its position vector.

- At every point along the path, the instantaneous velocity vector is tangent to the path at that point.



Velocity

v.得到

- Instantaneous Velocity ***v*** can be derived from ***r***, by taking the derivative of each component in Eq (3.1). $\vec{v} \Longleftrightarrow \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

n.导数，派生词

Each component of a particle's instantaneous velocity vector ...

$$v_x = \frac{dx}{dt} \qquad v_y = \frac{dy}{dt} \qquad v_z = \frac{dz}{dt} \tag{3.4}$$

... equals the instantaneous rate of change of its corresponding coordinate.

- Therefore, we get

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \tag{3.5}$$

or

$$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

- The magnitude of the instantaneous velocity vector

$$|\vec{v}| = v = \sqrt{v_x^2 + v_y^2 + v_z^2} \tag{3.6}$$

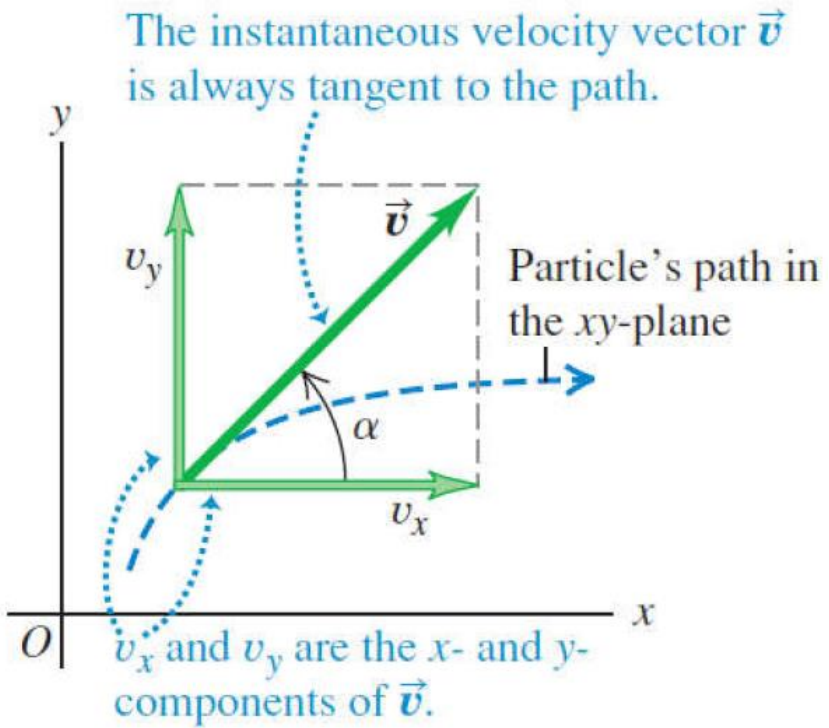
Velocity

- We use α for the direction of the instantaneous velocity vector to avoid confusion with the direction θ of the position vector of the particle.

For 2D motion

$$v = \sqrt{v_x^2 + v_y^2}$$

$$\tan\alpha = \frac{v_y}{v_x} \qquad (3.7)$$



Example for Velocity

Example 3.1 Calculating average and instantaneous velocity

A robotic vehicle, or rover, is exploring the surface of Mars. The stationary Mars lander is the origin of coordinates, and the surrounding Martian surface lies in the xy-plane. The rover, which we represent as a point, has x- and y-coordinates that vary with time:

$$x = 2.0\ \text{m} - (0.25\ \text{m/s}^2)t^2$$
$$y = (1.0\ \text{m/s})t + (0.025\ \text{m/s}^3)t^3$$

(0, 0)

(a) Find the rover's coordinates and distance from the lander at $t = 2.0$ s. (b) Find the rover's displacement and average velocity vectors for the interval $t = 0.0$ s to $t = 2.0$ s. (c) Find a general expression for the rover's instantaneous velocity vector $\vec{v}$. Express $\vec{v}$ at $t = 2.0$ s in component form and in terms of magnitude and direction.

Example for Velocity

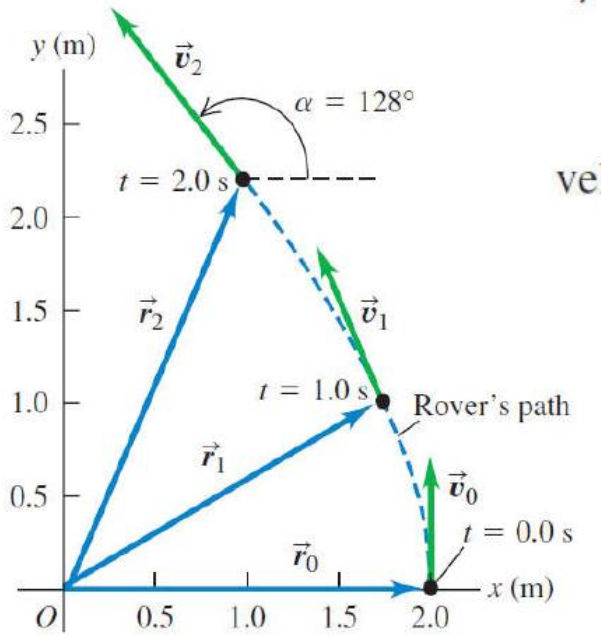
$$x = 2.0\ \text{m} - (0.25\ \text{m/s}^2)t^2$$
$$y = (1.0\ \text{m/s})t + (0.025\ \text{m/s}^3)t^3$$

EXECUTE: (a) At $t = 2.0$ s the rover's coordinates are

$$x = 2.0\ \text{m} - (0.25\ \text{m/s}^2)(2.0\ \text{s})^2 = 1.0\ \text{m}$$
$$y = (1.0\ \text{m/s})(2.0\ \text{s}) + (0.025\ \text{m/s}^3)(2.0\ \text{s})^3 = 2.2\ \text{m}$$

The rover's distance from the origin at this time is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.0\ \text{m})^2 + (2.2\ \text{m})^2} = 2.4\ \text{m}$$



(b) To find the displacement and average velocity over the given time interval, we first express

$$\vec{r} = x\hat{\imath} + y\hat{\jmath}$$
$$= [2.0\ \text{m} - (0.25\ \text{m/s}^2)t^2]\hat{\imath}$$
$$+ [(1.0\ \text{m/s})t + (0.025\ \text{m/s}^3)t^3]\hat{\jmath}$$

At $t = 0.0$ s the position vector $\vec{r}_0$ is

$$\vec{r}_0 = (2.0\ \text{m})\hat{\imath} + (0.0\ \text{m})\hat{\jmath}$$

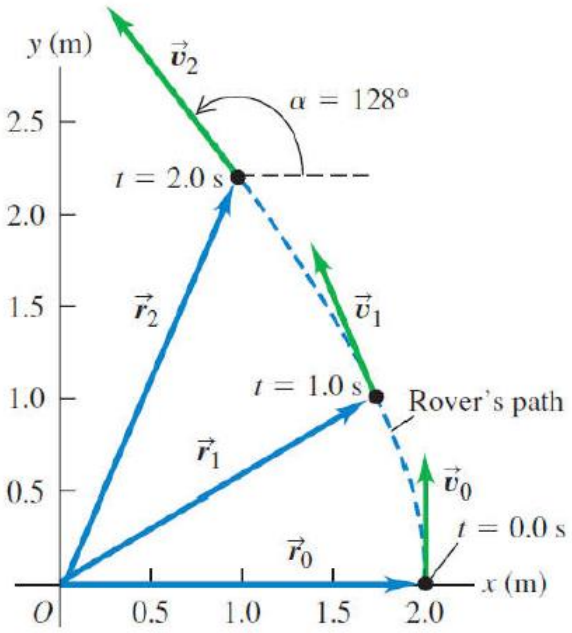
Example for Velocity

From part (a), the position vector $\vec{r}_2$ at $t = 2.0$ s is

$$\vec{r}_2 = (1.0\text{ m})\hat{i} + (2.2\text{ m})\hat{j}$$

The displacement from $t = 0.0$ s to $t = 2.0$ s is therefore

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_0 = (1.0\text{ m})\hat{i} + (2.2\text{ m})\hat{j} - (2.0\text{ m})\hat{i}$$
$$= (-1.0\text{ m})\hat{i} + (2.2\text{ m})\hat{j}$$



So

$$\vec{v}_{\text{av}} = \frac{\Delta\vec{r}}{\Delta t} = \frac{(-1.0\text{ m})\hat{i} + (2.2\text{ m})\hat{j}}{2.0\text{ s} - 0.0\text{ s}}$$
$$= (-0.50\text{ m/s})\hat{i} + (1.1\text{ m/s})\hat{j}$$

The components of this average velocity are

$$v_{\text{av-}x} = -0.50\text{ m/s}$$

$$v_{\text{av-}y} = 1.1\text{ m/s}$$

Example for Velocity

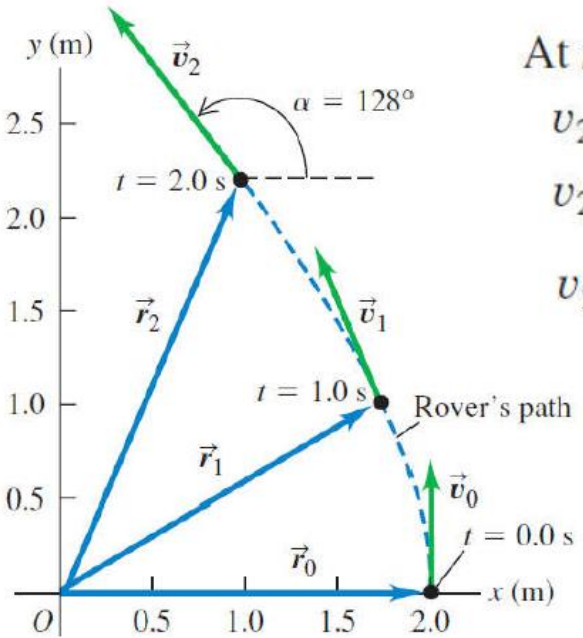
$$x = 2.0\ \text{m} - (0.25\ \text{m/s}^2)t^2$$
$$y = (1.0\ \text{m/s})t + (0.025\ \text{m/s}^3)t^3$$

(c) We can obtain

$$v_x = \frac{dx}{dt} = (-0.25\ \text{m/s}^2)(2t) \qquad v_y = \frac{dy}{dt} = 1.0\ \text{m/s} + (0.025\ \text{m/s}^3)(3t^2)$$

and

$$\vec{v} = v_x\hat{i} + v_y\hat{j} = (-0.50\ \text{m/s}^2)t\hat{i} + [1.0\ \text{m/s} + (0.075\ \text{m/s}^3)t^2]\hat{j}$$



At $t = 2.0$ s the velocity vector $\vec{v}_2$ has components

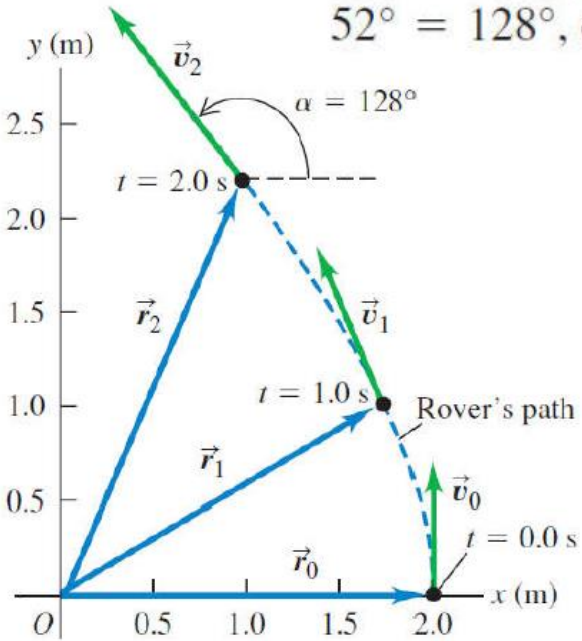
$$v_{2x} = (-0.50\ \text{m/s}^2)(2.0\ \text{s}) = -1.0\ \text{m/s}$$
$$v_{2y} = 1.0\ \text{m/s} + (0.075\ \text{m/s}^3)(2.0\ \text{s})^2 = 1.3\ \text{m/s}$$
$$v_2 = \sqrt{v_{2x}^2 + v_{2y}^2} = \sqrt{(-1.0\ \text{m/s})^2 + (1.3\ \text{m/s})^2}$$
$$= 1.6\ \text{m/s}$$

Example for Velocity

Figure 3.5 shows the direction of the velocity vector $\vec{v}_2$, which is at an angle α between 90° and 180° with respect to the positive x-axis. From Eq. (3.7) we have

$$\arctan\frac{v_y}{v_x} = \arctan\frac{1.3\ \text{m/s}}{-1.0\ \text{m/s}} = -52°$$

This is off by 180°; the correct value of the angle is $\alpha = 180° - 52° = 128°$, or 38° west of north.



Acceleration

- The average acceleration $\boldsymbol{a_{av}}$ of the car during this time interval is the velocity change divided by the time interval $t_2 - t_1 = \Delta t$

$$\vec{a}_{\text{av}} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t} \qquad (3.8)$$

- Instantaneous Acceleration

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \qquad (3.9)$$

$$\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k} \qquad (3.5)$$

$$\vec{a} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

➢ Its components are

Each component of a particle's instantaneous acceleration vector ...

$$a_x = \frac{dv_x}{dt} \qquad a_y = \frac{dv_y}{dt} \qquad a_z = \frac{dv_z}{dt} \qquad (3.10)$$

... equals the instantaneous rate of change of its corresponding velocity component.

Acceleration

- Instantaneous Acceleration can be derived from ***v*** $\vec{a} \Leftrightarrow \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt} \longrightarrow \vec{a} = \frac{d^2\vec{r}}{dt^2}$$

  - Its components are

$$\vec{a} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k} \quad (3.11)$$

向量形式化 x. v. a. 分支求解.

Since $v_x = \frac{dx}{dt}$ $v_y = \frac{dy}{dt}$ $v_z = \frac{dz}{dt}$

So

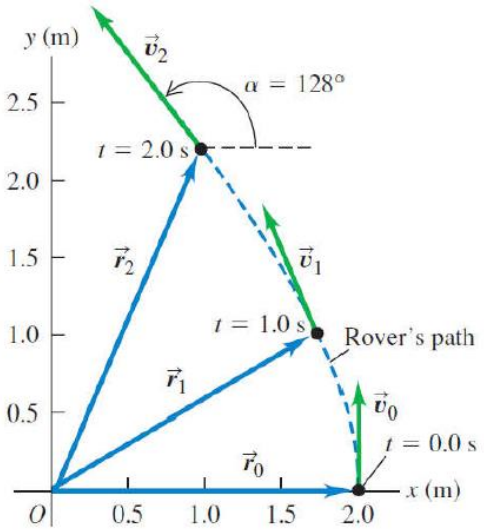
$$a_x = \frac{d^2x}{dt^2} \quad a_y = \frac{d^2y}{dt^2} \quad a_z = \frac{d^2z}{dt^2} \quad (3.12)$$

$$\vec{a} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k}$$

Example for Acceleration

Example 3.2

Let's return to the motions of the Mars rover in Example 3.1. (a) Find the components of the average acceleration for the interval $t = 0.0$ s to $t = 2.0$ s. (b) Find the instantaneous acceleration at $t = 2.0$ s.



EXECUTE: (a) In Example 3.1 we found that at $t = 0.0$ s the velocity components are

$$v_x = \frac{dx}{dt} = (-0.25\ \text{m/s}^2)(2t) \qquad v_y = \frac{dy}{dt} = 1.0\ \text{m/s} + (0.025\ \text{m/s}^3)(3t^2)$$

$$v_x = 0.0\ \text{m/s} \qquad v_y = 1.0\ \text{m/s}$$

and that at $t = 2.00$ s the components are

$$v_x = -1.0\ \text{m/s} \qquad v_y = 1.3\ \text{m/s}$$

Thus the components of average acceleration in the interval $t = 0.0$ s to $t = 2.0$ s are

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{-1.0\ \text{m/s} - 0.0\ \text{m/s}}{2.0\ \text{s} - 0.0\ \text{s}} = -0.50\ \text{m/s}^2$$

$$a_{\text{av-}y} = \frac{\Delta v_y}{\Delta t} = \frac{1.3\ \text{m/s} - 1.0\ \text{m/s}}{2.0\ \text{s} - 0.0\ \text{s}} = 0.15\ \text{m/s}^2$$

Example for Acceleration

Since $\vec{v} = v_x\hat{i} + v_y\hat{j} = (-0.50\ \text{m/s}^2)t\hat{i} + [1.0\ \text{m/s} + (0.075\ \text{m/s}^3)t^2]\hat{j}$

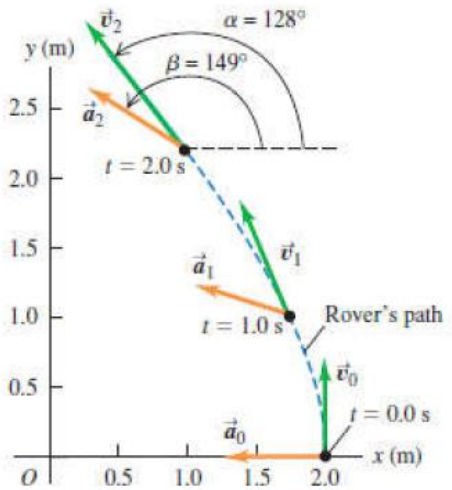
$$a_x = \frac{dv_x}{dt} = -0.50\ \text{m/s}^2 \qquad a_y = \frac{dv_y}{dt} = (0.075\ \text{m/s}^3)(2t)$$

$$\vec{a} = a_x\hat{i} + a_y\hat{j} = (-0.50\ \text{m/s}^2)\hat{i} + (0.15\ \text{m/s}^3)t\hat{j}$$

At $t = 2.0$ s, we can obtain

$$a_x = -0.50\ \text{m/s}^2 \qquad a_y = (0.15\ \text{m/s}^3)(2.0\ \text{s}) = 0.30\ \text{m/s}^2$$

$$\vec{a} = (-0.50\ \text{m/s}^2)\hat{i} + (0.30\ \text{m/s}^2)\hat{j}$$



The magnitude of acceleration at this time is

$$a = \sqrt{a_x^2 + a_y^2}$$

$$= \sqrt{(-0.50\ \text{m/s}^2)^2 + (0.30\ \text{m/s}^2)^2} = 0.58\ \text{m/s}^2$$

A sketch of this vector (Fig. 3.9) shows that the direction angle β of $\vec{a}$ with respect to the positive x-axis is between 90° and 180°. From Eq. (3.7) we have

$$\arctan\frac{a_y}{a_x} = \arctan\frac{0.30\ \text{m/s}^2}{-0.50\ \text{m/s}^2} = -31°$$

Hence $\beta = 180° + (-31°) = 149°$.

Acceleration on a curved path



? If a cyclist is going around a curve at constant speed, is he accelerating? If so, in which direction is he accelerating?

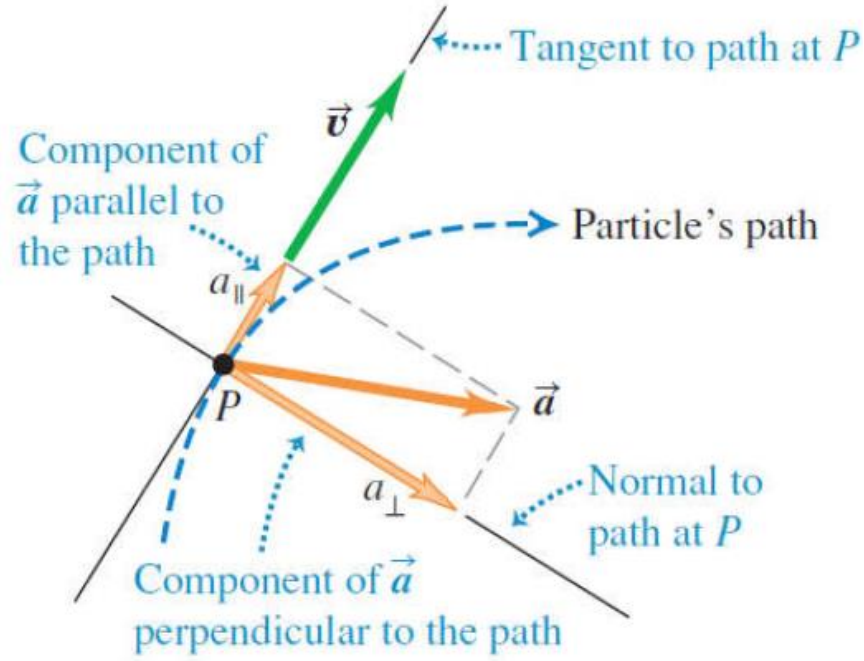
Parallel and Perpendicular Component of Acceleration

- Acceleration can be written in terms of two following components:

$$\vec{a} = \vec{a}_{\parallel} + \vec{a}_{\perp}$$

Where $\vec{a}_{\parallel}$ – parallel to the path, aka the parallel component

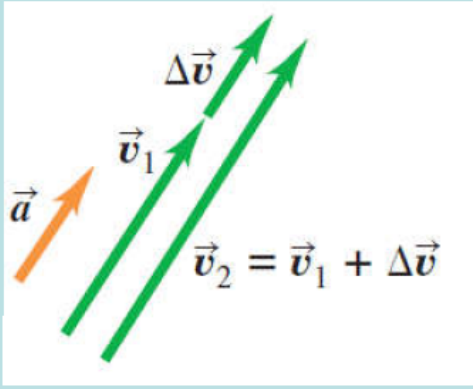
$\vec{a}_{\perp}$ – perpendicular to the path, aka the perpendicular component



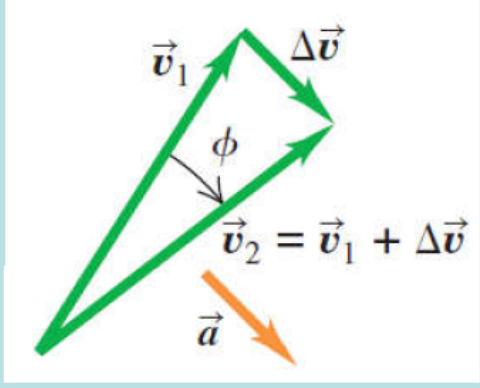
Parallel and Perpendicular Component of Acceleration

- The effect of $\vec{a}_{\parallel}$ and $\vec{a}_{\perp}$

$\vec{a}_{\parallel}$ — Changes only *magnitude* of velocity: speed changes; direction doesn't.



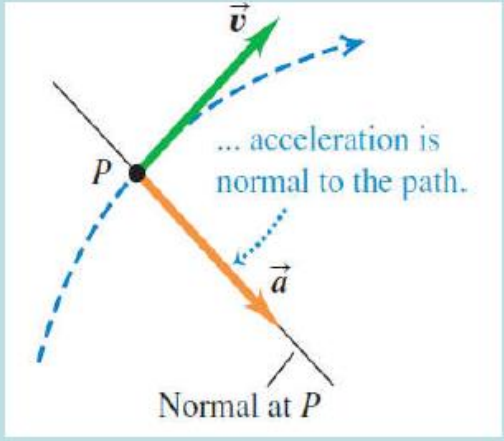
$\vec{a}_{\perp}$ — Changes only *direction* of velocity: magnitude doesn't change, particle follows curved path at constant speed.



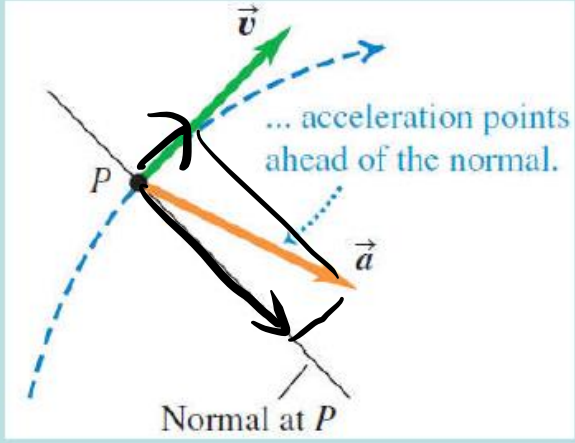
Parallel and Perpendicular Component of Acceleration

- Velocity and acceleration vectors for a particle moving through a point P on a curved path

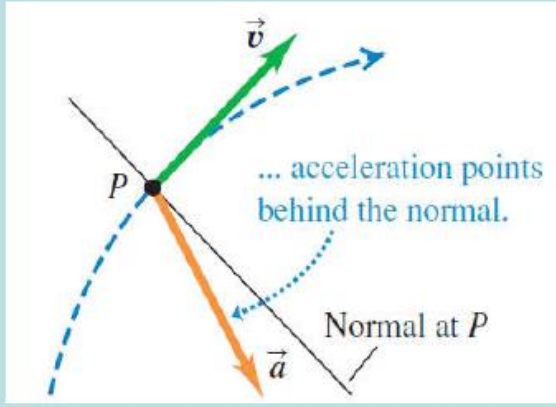
(a) When speed is constant along a curved path.



(b) When speed is increasing along a curved path.

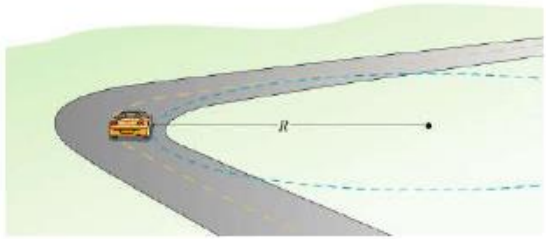


(c) When speed is decreasing along a curved path

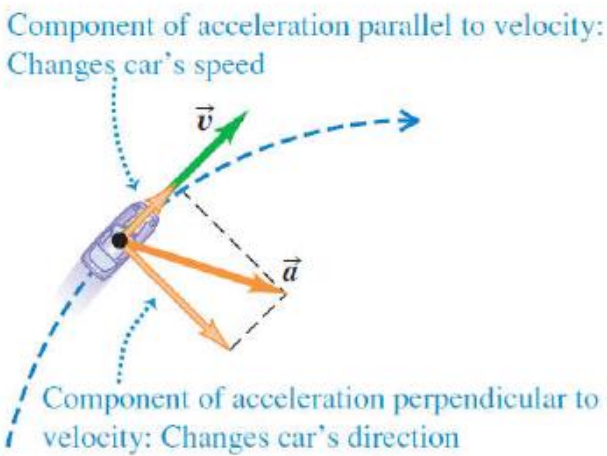


Parallel and Perpendicular Component of Acceleration

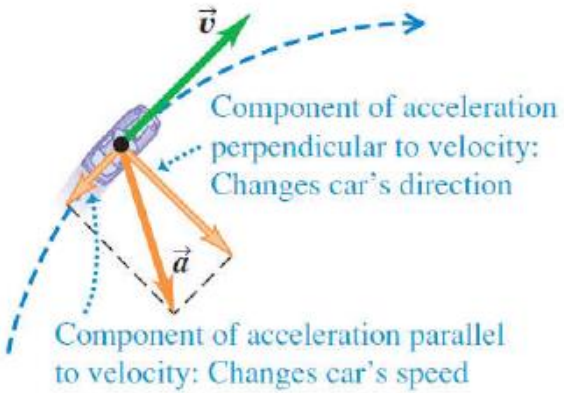
- Example: a car moving along a circle path



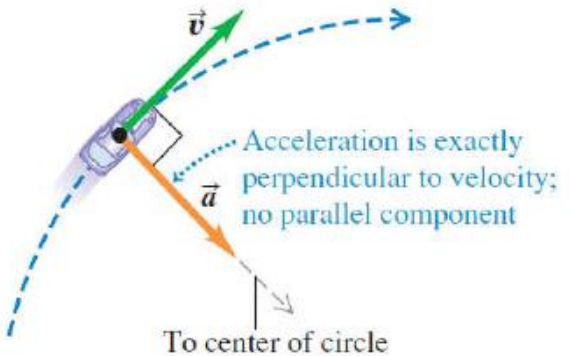
(a) Car speeding up along a circular path



(b) Car slowing down along a circular path



(c) Uniform circular motion: Constant speed along a circular path



Parallel and Perpendicular Component of Acceleration

- A car speeding up a circular path



Example of Acceleration

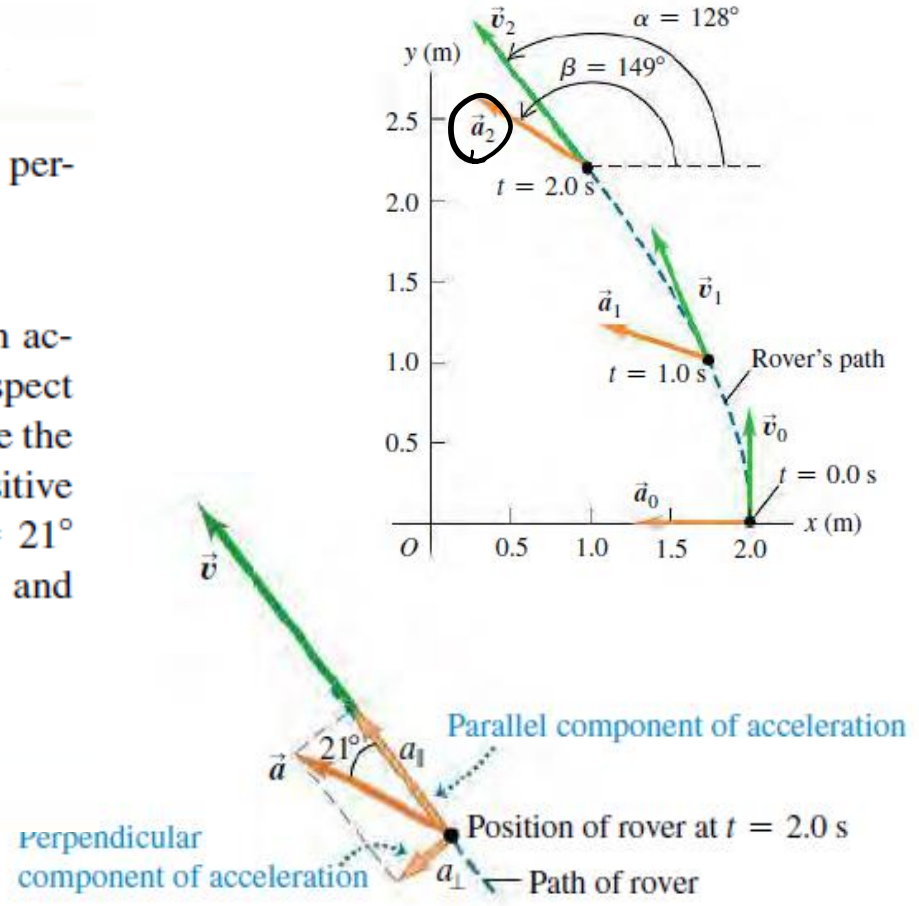
EXAMPLE 3.3

For the rover of Examples 3.1 and 3.2, find the parallel and perpendicular components of the acceleration at $t = 2.0$ s.

EXECUTE: From Example 3.2, at $t = 2.0$ s the particle has an acceleration of magnitude 0.58 m/s^2 at an angle of 149° with respect to the positive x-axis. In Example 3.1 we found that at this time the velocity vector is at an angle of 128° with respect to the positive x-axis. The angle between $\vec{a}$ and $\vec{v}$ is therefore 149° − 128° = 21° (**Fig. 3.13**). Hence the components of acceleration parallel and perpendicular to $\vec{v}$ are

$$a_{\parallel} = a\cos 21° = (0.58\ \text{m/s}^2)\cos 21° = 0.54\ \text{m/s}^2$$

$$a_{\perp} = a\sin 21° = (0.58\ \text{m/s}^2)\sin 21° = 0.21\ \text{m/s}^2$$



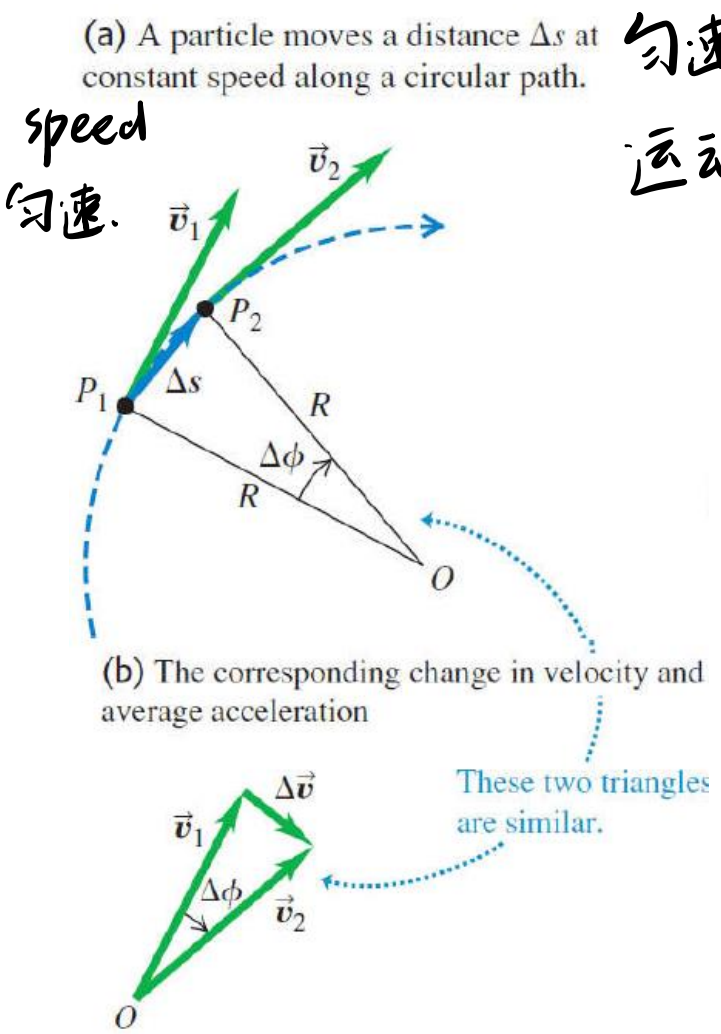
Motion in a Circle – Uniform Circular Motion

- The two triangles are similar.
- Hence, ratios of corresponding sides of similar triangles are equal:

$$\frac{|\Delta\vec{v}|}{v_1} = \frac{\Delta s}{R} \quad \text{or} \quad |\Delta\vec{v}| = \frac{v_1}{R}\Delta s$$

- The magnitude a_{av} of the average acceleration during Δt is therefore

$$a_{av} = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{v_1}{R}\frac{\Delta s}{\Delta t}$$

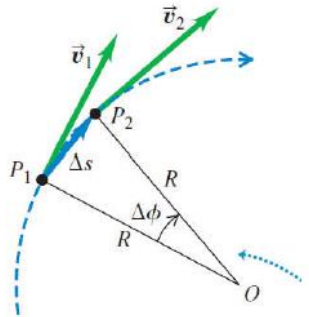


Motion in a Circle – Uniform Circular Motion

- The magnitude a of the instantaneous acceleration $\boldsymbol{a}$ at point P_1 is the limit of this expression as we take point P_2 closer and closer to point P_1:

$$a = \lim_{\Delta t \to 0} \frac{v_1}{R}\frac{\Delta s}{\Delta t} = \frac{v_1}{R}\lim_{\Delta t \to 0}\frac{\Delta s}{\Delta t}$$

The limit of $\Delta s/\Delta t$ is the speed v_1 at point P_1. v_1

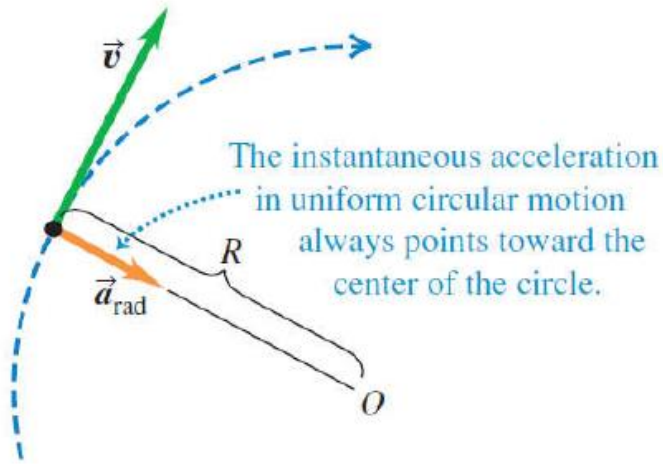


(c) The instantaneous acceleration

- Let v represent the speed at any point:

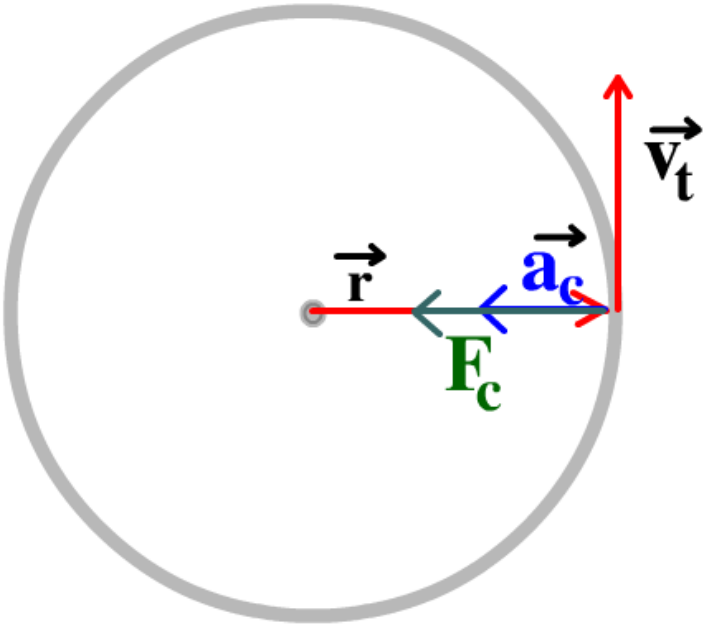
Magnitude of acceleration of an object in **uniform circular motion**: $a_{\text{rad}} = \frac{v^2}{R}$ (3.27)

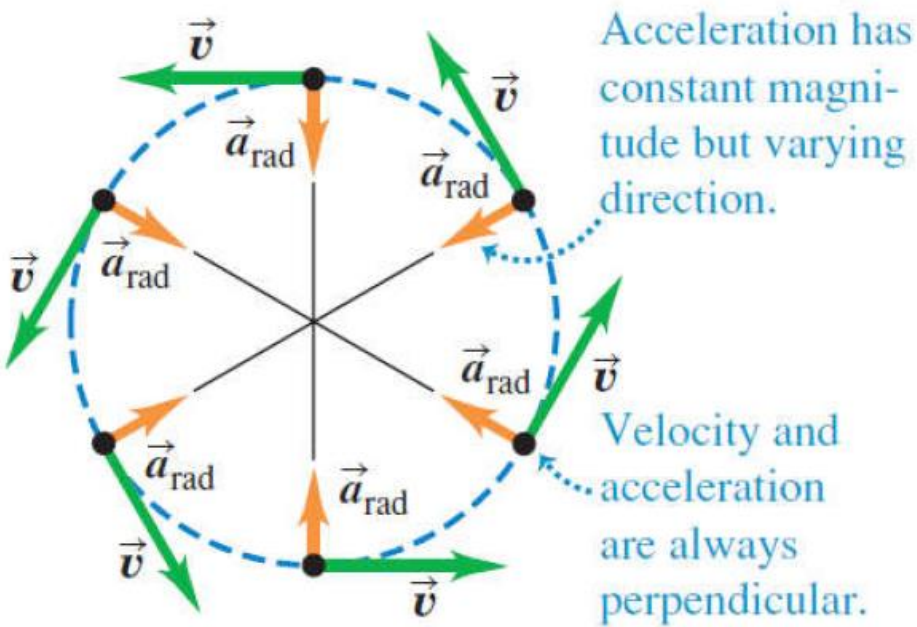
v: Speed of object; R: Radius of object's circular path



- With its direction perpendicular to velocity v and inward along the radius, aka centripetal acceleration.

Motion in a Circle – Uniform Circular Motion





- Expression in terms of period T, substitute Eq (3.28) into Eq (3.27)

$$v = \frac{2\pi R}{T} \qquad (3.28)$$

$$a_{rad} = \frac{4\pi^2 R}{T^2} \qquad (3.29)$$

$\leftarrow a = \frac{v^2}{R}$

Example of Uniform Circular Motion

Example 3.12

Passengers on a carnival ride move at constant speed in a horizontal circle of radius 5.0 m, making a complete circle in 4.0 s. What is their acceleration? R=5.

D = 直径 ⇒ diameter.

Solution: $T = 4.0\ \text{s}$

$$a_{\text{rad}} = \frac{4\pi^2 R}{T^2}$$

$a = \frac{v^2}{R}$ $\quad v = \omega r$

$$a_{\text{rad}} = \frac{4\pi^2(5.0\ \text{m})}{(4.0\ \text{s})^2} = 12\ \text{m/s}^2 = 1.3g$$

$T = \frac{2\pi}{\omega}$

or

$$v = \frac{2\pi R}{T} = \frac{2\pi(5.0\ \text{m})}{4.0\ \text{s}} = 7.9\ \text{m/s}$$

$$a_{\text{rad}} = \frac{v^2}{R} = \frac{(7.9\ \text{m/s})^2}{5.0\ \text{m}} = 12\ \text{m/s}^2$$

Nonuniform Circular Motion

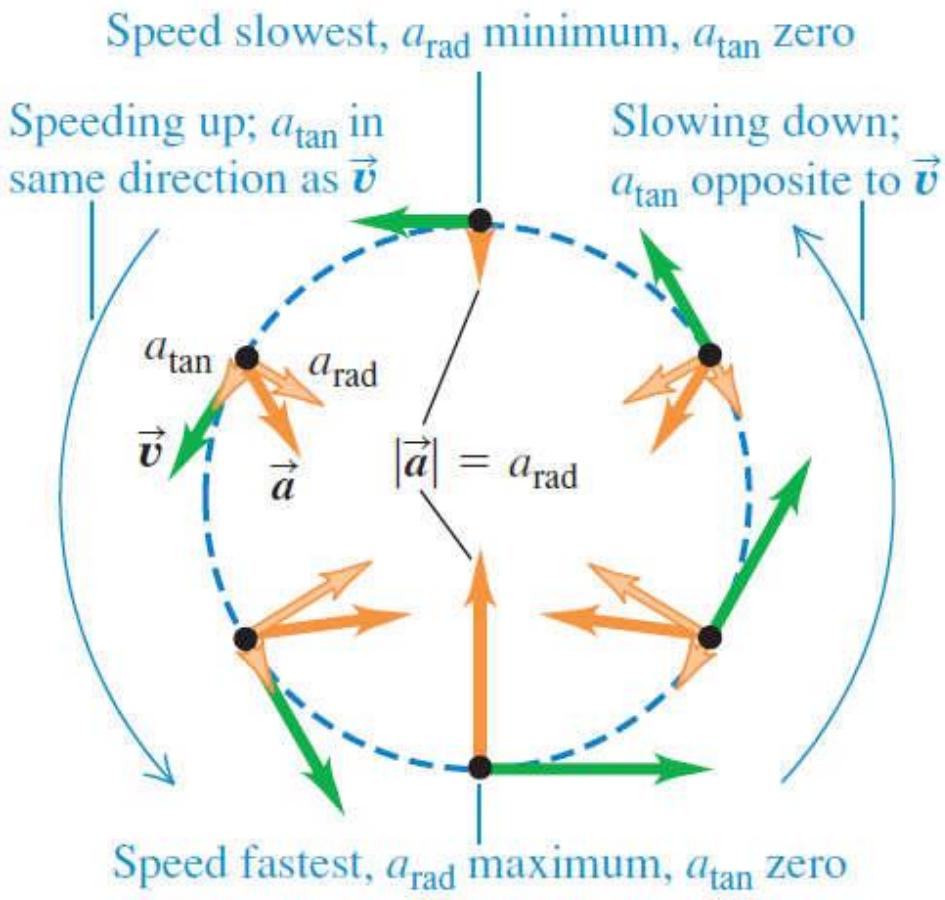
- The particle's speed is not constant as it goes around the circle.
  - Two components:

$$a_{\text{rad}} = \frac{v^2}{R}$$

and

$$a_{\text{tan}} = \frac{d|\vec{v}|}{dt} \qquad (3.30)$$

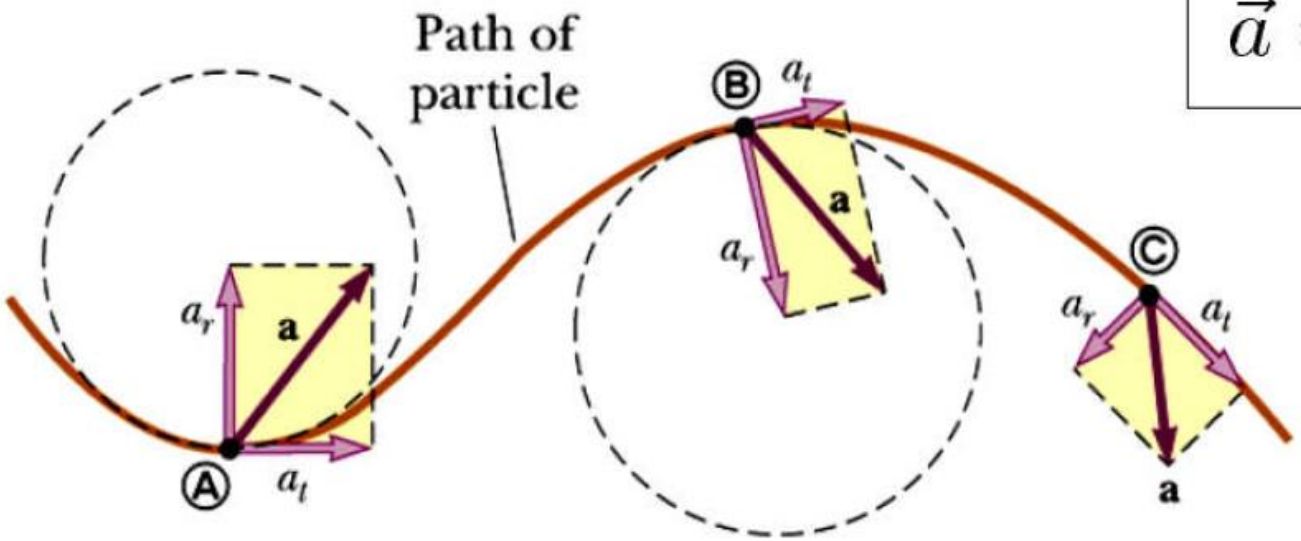
- The tangential component is in the same direction as the velocity.



Tangential and Radial Acceleration

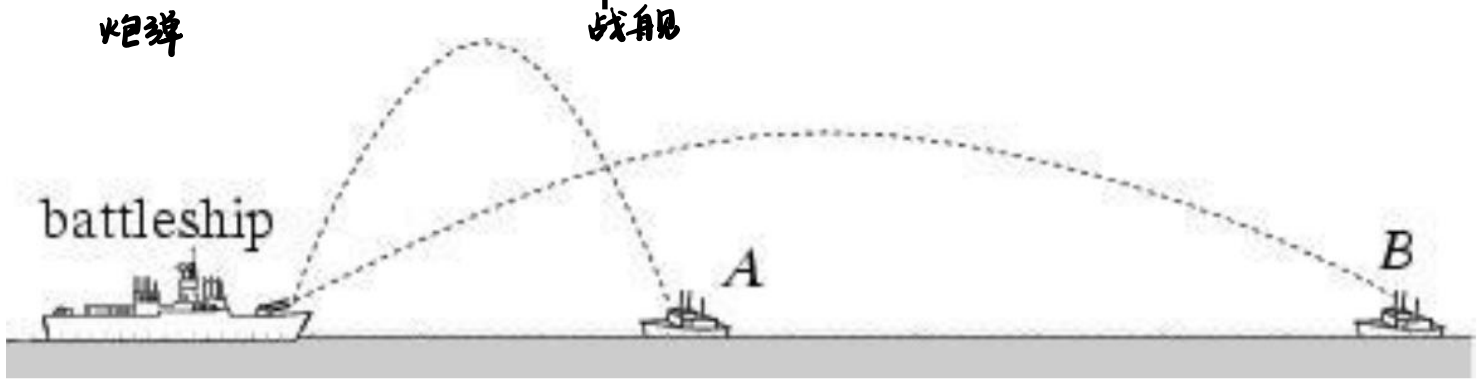
- In general, for any motion, where there is a change in the direction and magnitude of velocity, the acceleration is the vector sum of the tangential and radial accelerations:

$$\vec{a} = \vec{a}_r + \vec{a}_t$$



Projectile Motion

- A projectile is any body that is given an initial velocity and then follows a path determined entirely by the effects of **gravitational acceleration** and **air resistance**.
  - A thrown ball
  - A bullet from a rifle
  - A shell from a warship

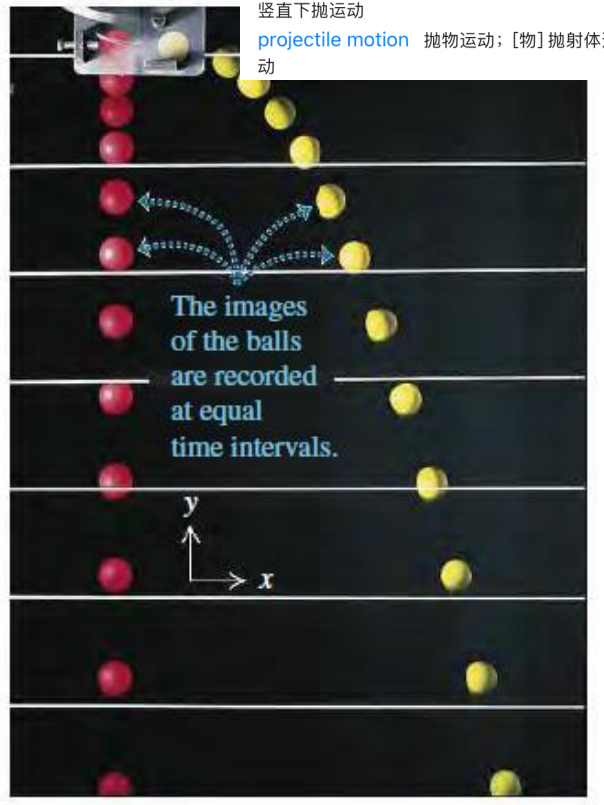


projectile motions 弹丸运动
horizontal projectile motion 平抛运动
oblique projectile motion 斜抛运动
hyperbolic projective motion 双曲射影运动
flat projectile motion 平抛运动
vertically upward projectile motion 竖直上抛运动
vertically downword projectile motion 竖直下抛运动
projectile motion 抛物运动；[物] 抛射体运动

Projectile Motion 抛物运动

An Experiment on Projectile Motion

- Red ball and yellow ball released from the same height simultaneously, but yellow ball with an initial velocity in the horizontal direction.
- Experiment results: two balls touch the ground at the same time.
  - ➢ Horizontal motion has no effect on its vertical motion
  - ➢ Projectile motion is a combination of horizontal motion with a constant velocity and a vertical motion with an constant acceleration.



Projectile Motion

n.发射体；抛射物

- A projectile moves in a vertical plane that contains the initial velocity vector $\boldsymbol{v_0}$.
- Its trajectory depends only on $\boldsymbol{v_0}$ and on the downward acceleration due to gravity.

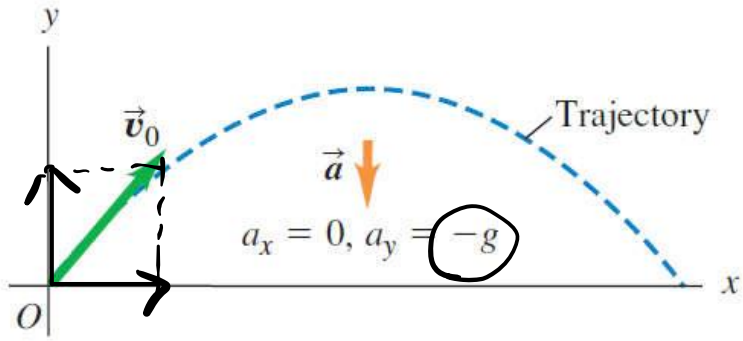
n.弹道，轨迹

$$\vec{a} = \vec{a}_x + \vec{a}_y = a_x \hat{i} + a_y \hat{j}$$

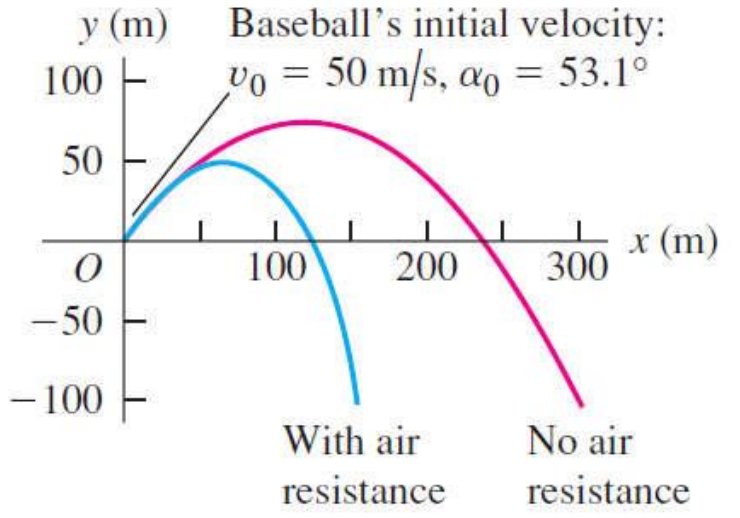
with

$$a_x = 0,\ a_y = -g \qquad (3.13)$$

- No air resistance (implied or specially emphasized in the question)

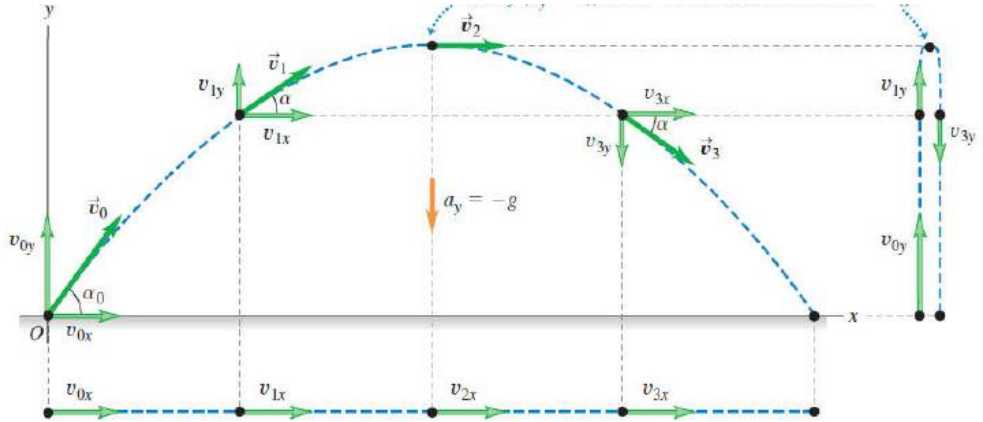


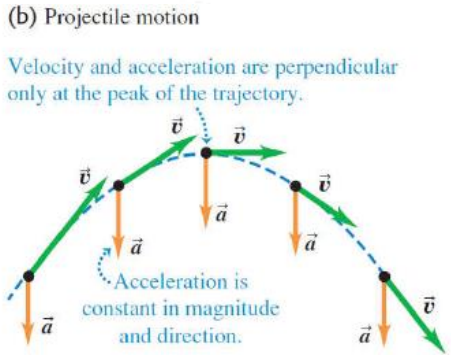
If air resistance cannot be ignored, the height and distance of a projectile will be decreased



Projectile Motion

- Horizontal (x-direction): constant-velocity motion, $a_x = 0$
- Perpendicular (y-direction): constant-acceleration motion, $a_y = -g$





- Suppose that at time t = 0 our particle is at the point (x_0, y_0) and that at this time its velocity components have the initial values v_{0x} and v_{0y}.

x-direction

$$v_x = v_{0x} \quad (3.14)$$

$$x = x_0 + v_{0x}t \quad (3.15)$$

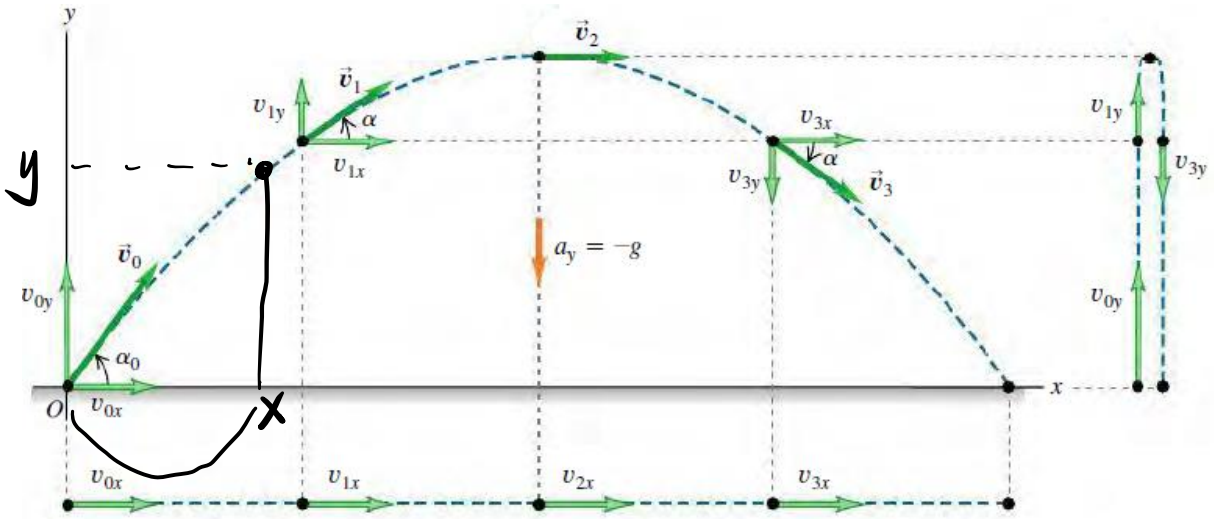
y-direction

$$v_y = v_{0y} - gt \quad (3.16)$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2 \quad (3.17)$$

*Recall Eq (2.8) and (2.12).

Projectile Motion



— Initial value for velocity

$$\begin{cases} v_{0x} = v_0 \cos\alpha_0 \\ v_{0y} = v_0 \sin\alpha_0 \end{cases} \quad (3.18)$$

By setting $x_0 = y_0 = 0$, we have

$$x = (v_0 \cos\alpha_0)t \quad (3.19)$$

$$y = (v_0 \sin\alpha_0)t - \tfrac{1}{2}gt^2 \quad (3.20)$$

$$v_x = v_0 \cos\alpha_0 \quad (3.21)$$

$$v_y = v_0 \sin\alpha_0 - gt \quad (3.22)$$

- We can derive an equation for the trajectory's shape in terms of *x* and *y* by eliminating *t*.
- From Eqs (3.19) and (3.20), we find $t = x/(v_0 \cos\alpha_0)$ and we get a parabolic equation:

$$y = (\tan\alpha_0)x - \frac{g}{2v_0^2 \cos^2\alpha_0}x^2 \quad (3.26) \quad \Longleftrightarrow \quad y = bx - cx^2$$

- In a simple model of projectile motion, the trajectory is always a parabola.

Projectile Motion

- Maximum height can be derived by the following equation:

$$v_{0y} = v_0 \sin\alpha_0$$

$$v_y^2 = v_{0y}^2 - 2g\,(y - y_0)$$

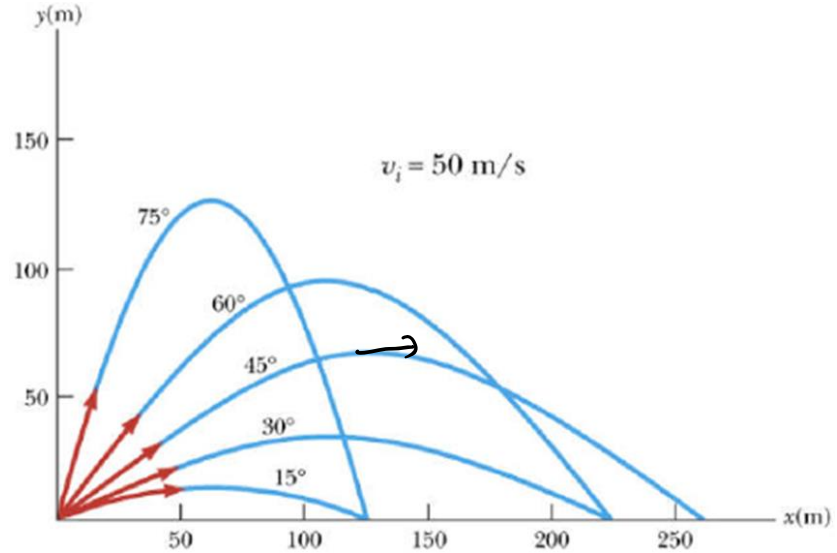
- The highest point in any trajectory, called the apex, is reached when $v_y = 0$. With $y_0 = 0$, we get:

$$h = y_{\max} = \frac{v_0^2 \sin^2 \alpha_0}{2g}$$

- The range *R* of a projectile on level ground for which air resistance is negligible is given by:

$$y = (\tan\alpha_0)x - \frac{g}{2v_0^2\cos^2\alpha_0}x^2 \qquad R = x = \frac{v_0^2 \sin(2\alpha_0)}{g}$$

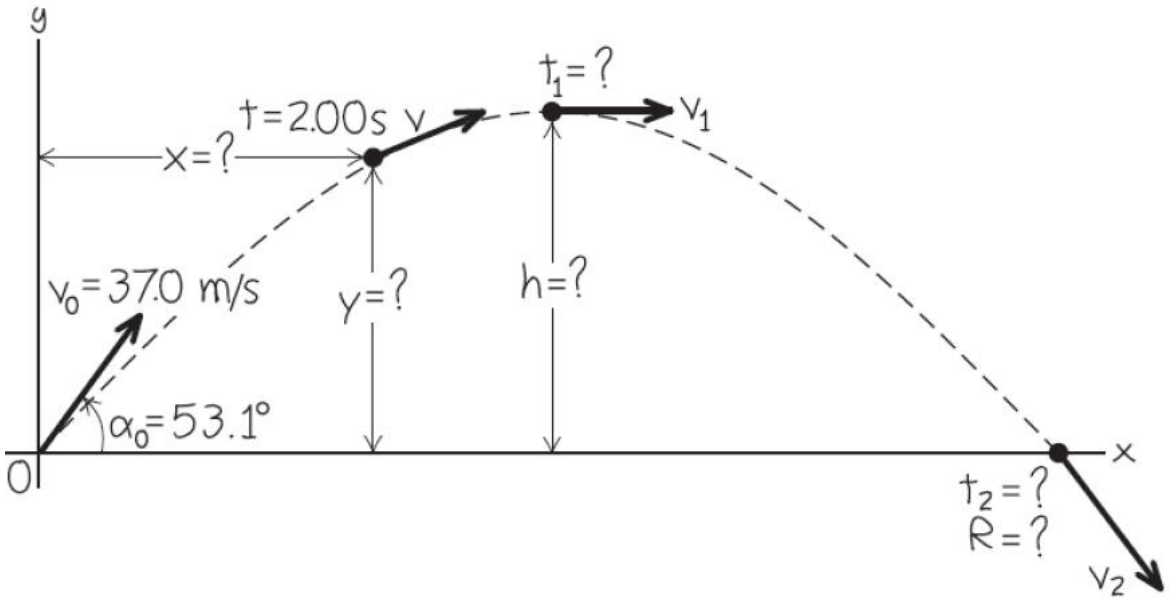
- *R* is maximum when $\alpha_0 = 45°$, if air resistance is neglected.



Example for Projectile Motion

Example 3.7

A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0$ m/s at an angle $\alpha_0 = 53.1°$. (a) Find the position of the ball and its velocity (magnitude and direction) at $t = 2.00$ s. (b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the *horizontal range* R—that is, the horizontal distance from the starting point to where the ball hits the ground.



Example for Projectile Motion

Solution: (a) The components of the initial value are

$$v_{0x} = v_0 \cos\alpha_0 = (37.0\ \text{m/s})\cos 53.1^\circ = 22.2\ \text{m/s}$$

$$v_{0y} = v_0 \sin\alpha_0 = (37.0\ \text{m/s})\sin 53.1^\circ = 29.6\ \text{m/s}$$

At t = 2.0s,

$$x = v_{0x}t = (22.2\ \text{m/s})(2.00\ \text{s}) = 44.4\ \text{m}$$

$$y = v_{0y}t - \tfrac{1}{2}gt^2 = (29.6\ \text{m/s})(2.00\ \text{s}) - \tfrac{1}{2}(9.80\ \text{m/s}^2)(2.00\ \text{s})^2 = 39.6\ \text{m}$$

$$v_x = v_{0x} = 22.2\ \text{m/s}$$

$$v_y = v_{0y} - gt = 29.6\ \text{m/s} - (9.80\ \text{m/s}^2)(2.00\ \text{s}) = 10.0\ \text{m/s}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(22.2\ \text{m/s})^2 + (10.0\ \text{m/s})^2} = 24.4\ \text{m/s}$$

The *y*-component of velocity is positive at t = 2.00 s, so the ball is still moving upward

$$\alpha = \arctan\left(\frac{10.0\ \text{m/s}}{22.2\ \text{m/s}}\right) = \arctan 0.450 = 24.2^\circ$$

The ball is moving at 24.4 m/s in a direction 24.2° above the horizontal.

(b) At the highest point, the vertical velocity v_y is zero. Call the time when this happens t_1; then

$$v_y = v_{0y} - gt_1 = 0$$

$$t_1 = \frac{v_{0y}}{g} = \frac{29.6\ \text{m/s}}{9.80\ \text{m/s}^2} = 3.02\ \text{s}$$

Example for Projectile Motion

Solution(cont'): The height h at the highest point is the value of y at time t_1:

$$h = v_{0y}t_1 - \tfrac{1}{2}gt_1^2 = (29.6\ \text{m/s})(3.02\ \text{s}) - \tfrac{1}{2}(9.80\ \text{m/s}^2)(3.02\ \text{s})^2 = 44.7\ \text{m}$$

(c) We'll find the horizontal range in two steps. First, we find the time t_2 when $y = 0$ (the ball is at ground level):

$$y = 0 = v_{0y}t_2 - \tfrac{1}{2}gt_2^2 = t_2\left(v_{0y} - \tfrac{1}{2}gt_2\right)$$

$$h = y_{\max} = \frac{v_0^2 \sin^2 \alpha_0}{2g}$$

This is a quadratic equation for t_2. It has two roots:

$$t_2 = 0 \quad \text{and} \quad t_2 = \frac{2v_{0y}}{g} = \frac{2(29.6\ \text{m/s})}{9.80\ \text{m/s}^2} = 6.04\ \text{s}$$

The ball is at $y = 0$ at both times. The ball *leaves* the ground at $t_2 = 0$, and it hits the ground at $t_2 = 2v_{0y}/g = 6.04$ s. The horizontal range R is the value of x when the ball returns to the ground at $t_2 = 6.04$ s:

$$R = x = \frac{v_0^2 \sin(2\alpha_0)}{g}$$

$$R = v_{0x}t_2 = (22.2\ \text{m/s})(6.04\ \text{s}) = 134\ \text{m}$$

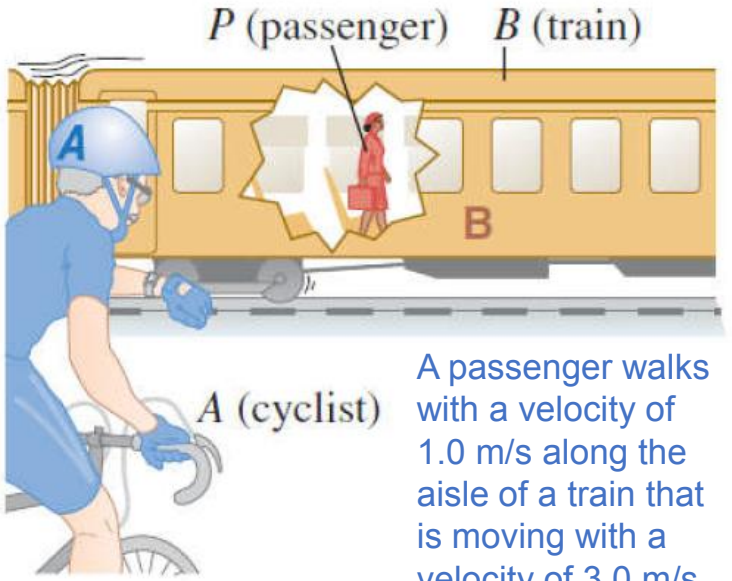
The vertical component of velocity when the ball hits the

$$v_y = v_{0y} - gt_2 = 29.6\ \text{m/s} - (9.80\ \text{m/s}^2)(6.04\ \text{s}) = -29.6\ \text{m/s}$$

That is, v_y has the same magnitude as the initial vertical velocity v_{0y} but the opposite direction (down). Since v_x is constant, the angle $\alpha = -53.1°$ (below the horizontal) at this point is the negative of the initial angle $\alpha_0 = 53.1°$.

Relative Velocity 相对速度

- Each observer, equipped in principle with a meter stick and a stopwatch, forms what we call a frame of reference.
- A: the cyclist's frame of reference (at rest with respect to ground)
- B: the frame of reference of the moving train
- Observers in different frames of reference may describe the motion of an object differently (in terms of displacement, velocity, acceleration…)



A passenger walks with a velocity of 1.0 m/s along the aisle of a train that is moving with a velocity of 3.0 m/s.

1. Passenger (P) / B(train) —1.0 m/s

2. Train (B) / Cyclist (A) —3.0 m/s

Passenger (P) / Cyclist（A） 4 ?

Relative Velocity

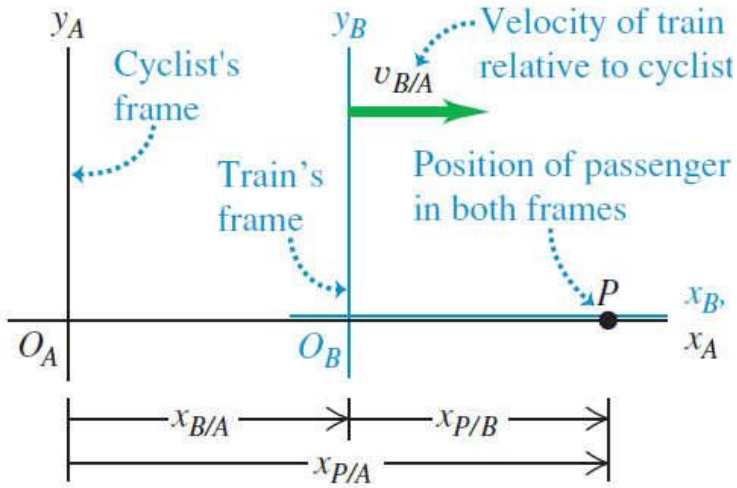
- Relative Position in 1D
  - The Position P with respect to A: $x_{P/A} = x_{P/B} + x_{B/A}$ (3.31)
- Derivative of $X_{P/A}$ with respect to time

$$\frac{dx_{P/A}}{dt} = \frac{dx_{P/B}}{dt} + \frac{dx_{B/A}}{dt}$$

  - Gives the Relative Velocity in 1D, i.e., the x-velocity of P relative to A:

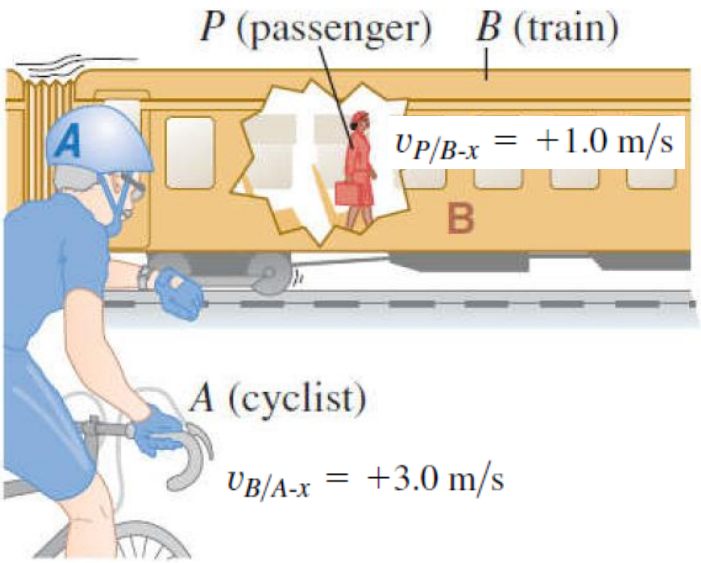
Relative velocity along a line: $v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x}$ (3.32)

- $v_{P/A\text{-}x}$: x-velocity of P relative to A
- $v_{P/B\text{-}x}$: x-velocity of P relative to B
- $v_{B/A\text{-}x}$: x-velocity of B relative to A



Relative Velocity

- Therefore, the passenger's velocity $v_{P/A\text{-}x}$ relative to the cyclist is



$$v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x}$$

$$v_{P/A\text{-}x} = +1.0 \text{ m/s} + 3.0 \text{ m/s} = +4.0 \text{ m/s}$$

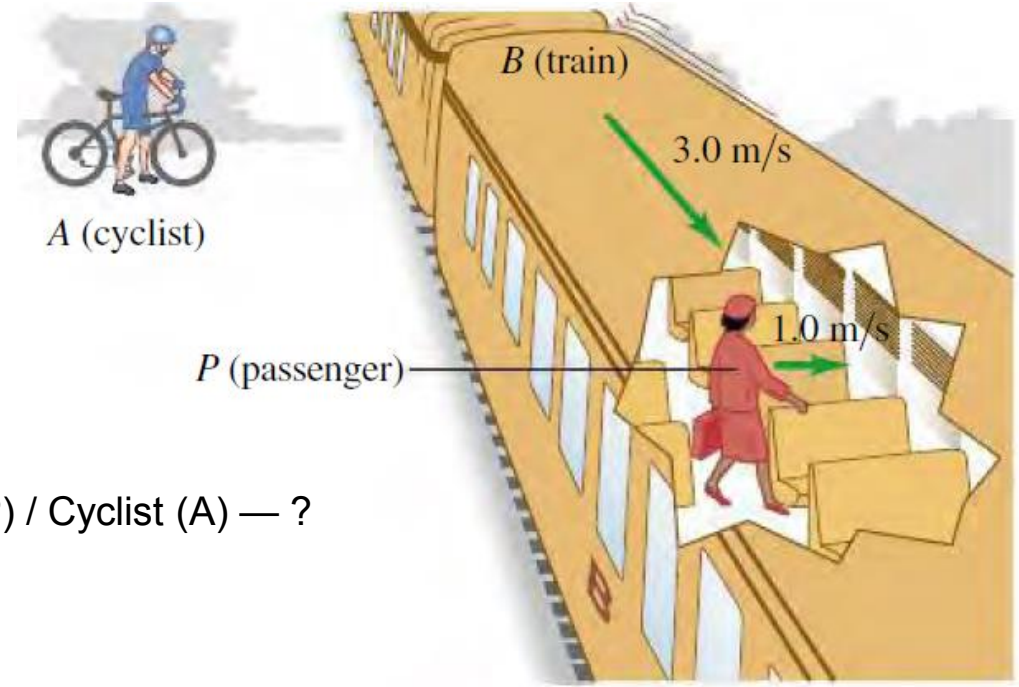
- When the passenger looks out the window, the stationary cyclist on the ground appears to her to be moving backward; we call the cyclist's velocity relative to her $v_{A/P\text{-}x}$.
- This is just the negative of the *passenger's* velocity relative to the *cyclist*, $v_{P/A\text{-}x}$.
- In general, if *A* and *B* are any two points or frames of reference,

$$v_{A/B\text{-}x} = -v_{B/A\text{-}x} \qquad (3.33)$$

Relative Velocity

- Relative Velocity in 2D or 3D

  1. Passenger (P) / B(train) —1.0 m/s, but in different direction.
  2. Train (B) / Cyclist (A) — 3.0 m/s,

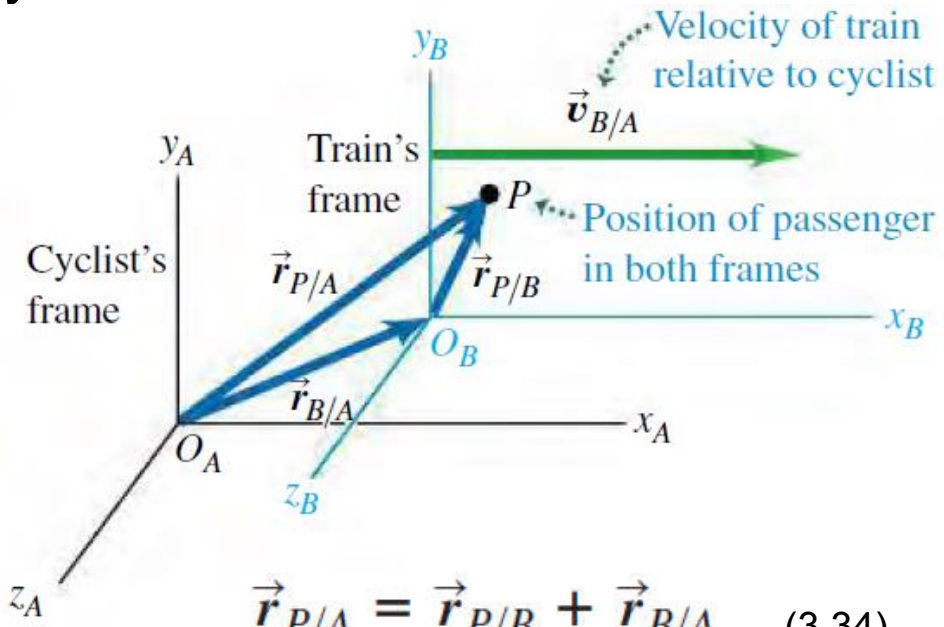


Passenger (P) / Cyclist (A) — ?

$\sqrt{10}$ m/s

Relative Velocity

- Relative Velocity in 2D or 3D



$$\vec{r}_{P/A} = \vec{r}_{P/B} + \vec{r}_{B/A} \qquad (3.34)$$

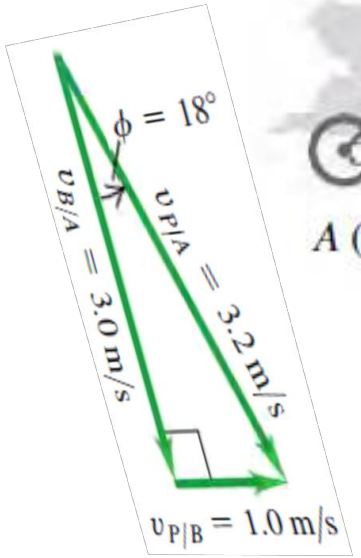
Relative velocity in space:

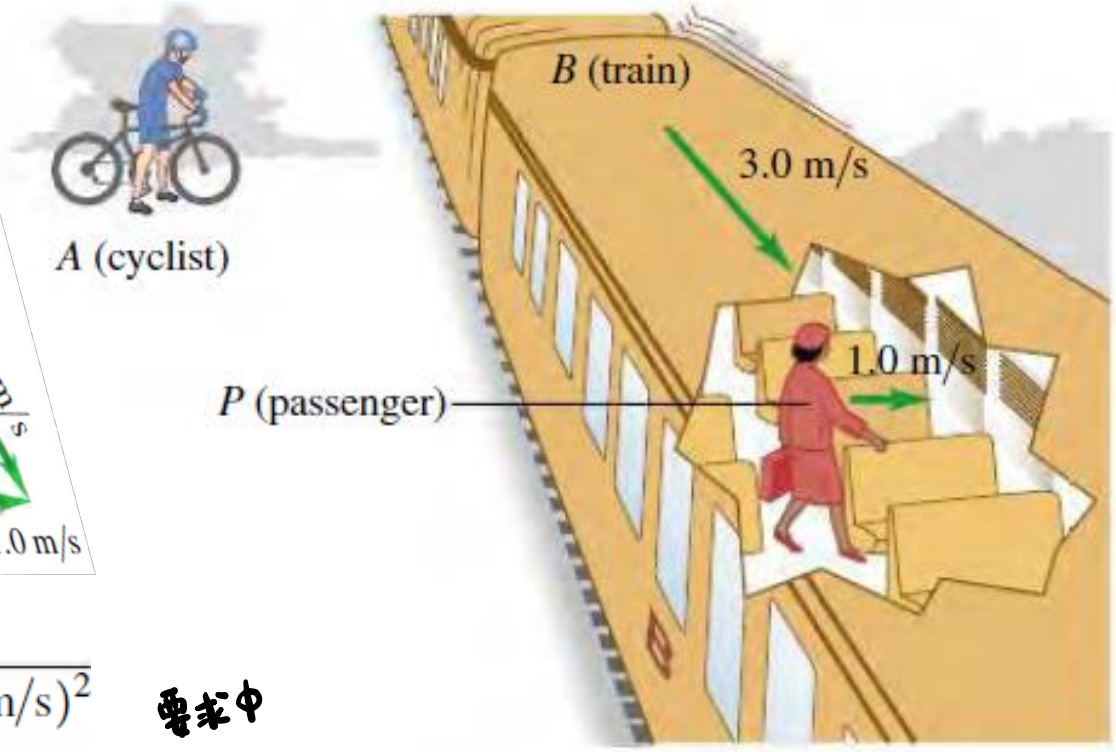
$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A} \qquad (3.35)$$

Velocity of P relative to A | Velocity of P relative to B | Velocity of B relative to A

Relative Velocity

- Relative Velocity in 2D or 3D





$$v_{P/A} = \sqrt{(3.0\ \text{m/s})^2 + (1.0\ \text{m/s})^2}$$
$$= \sqrt{10\ \text{m}^2/\text{s}^2} = 3.2\ \text{m/s}$$

要求 ϕ

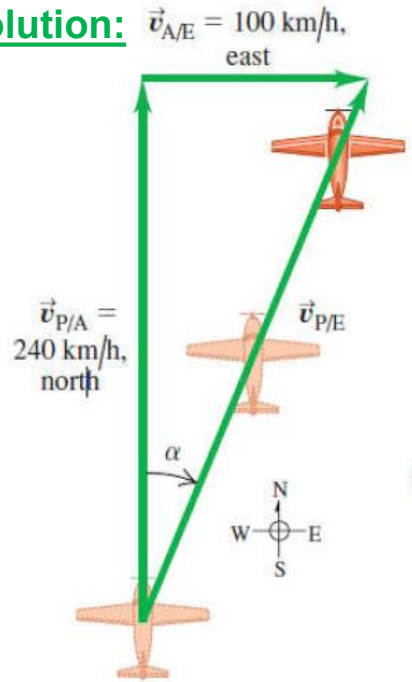
$$\tan\phi = \frac{v_{P/B}}{v_{B/A}} = \frac{1.0\ \text{m/s}}{3.0\ \text{m/s}} \quad \text{and} \quad \phi = 18°$$

Example for Relative Velocity

Example 3.14

An airplane's compass indicates that it is headed due north, and its airspeed indicator shows that it is moving through the air at 240 km/h. If there is a 100-km/h wind from west to east, what is the velocity of the airplane relative to the earth?

Solution:



$$\vec{v}_{P/A} = 240 \text{ km/h} \quad \text{due north}$$
$$\vec{v}_{A/E} = 100 \text{ km/h} \quad \text{due east}$$

$$\vec{v}_{P/E} = \vec{v}_{P/A} + \vec{v}_{A/E}$$

$$v_{P/E} = \sqrt{(240 \text{ km/h})^2 + (100 \text{ km/h})^2} = 260 \text{ km/h}$$

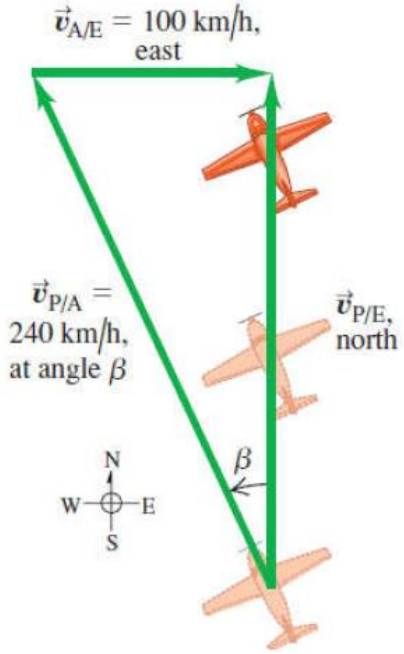
$$\alpha = \arctan\left(\frac{100 \text{ km/h}}{240 \text{ km/h}}\right) = 23^\circ \text{ E of N}$$

Example for Relative Velocity

Example 3.15

With wind and airspeed as in Example 3.14, in what direction should the pilot head to travel due north? What will be her velocity relative to the earth?

Solution:



$\vec{v}_{P/E}$ = magnitude unknown due north

$\vec{v}_{P/A}$ = 240 km/h direction unknown

$\vec{v}_{A/E}$ = 100 km/h due east

$$\vec{v}_{P/E} = \vec{v}_{P/A} + \vec{v}_{A/E}$$

$$v_{P/E} = \sqrt{(240\ \text{km/h})^2 - (100\ \text{km/h})^2} = 218\ \text{km/h}$$

$$\beta = \arcsin\left(\frac{100\ \text{km/h}}{240\ \text{km/h}}\right) = 25°$$

Summary

- 2D and 3D Motion are discussed and studied with examples.
  - Physical quantities in 2D & 3D motion (Position Vector, Velocity, Acceleration)
  - 2 typical 2D motions (1. Projectile motion. 2. Motion in a circle.)
  - Relative velocity
  - Though vectors are used for physical quantities studied in 2D & 3D motion, what we mainly copy with are their (x-, y-, or z-) components, which are all **scalars**.
- Textbook: Chapter 3 2D and 3D Motion